

Erik Hogset

Mechanical Engineering Senior | Design, Energy Systems, and Manufacturing

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Portfolio: erikhogset.github.io

Education

Boston University College of Engineering, Boston, MA

Expected May 2026

B.S. in Mechanical Engineering

Relevant Coursework: Heat Transfer • Fluid Mechanics • Mechanics of Materials • Electromechanical Design • Automation & Manufacturing Methods • Additive Manufacturing • Instrumentation • Introduction to Clean Energy • Senior Design

Technical Skills

Manufacturing: CNC Milling, CNC Turning, Manual Milling, Manual Turning, 3D Printing (FFF, SLA), TIG Welding, CO₂ Laser Cutter, Waterjet Cutting, Sheet Metal Fabrication

CAD / CAM: SolidWorks, Autodesk Fusion 360, Onshape

Programming / Controls: MATLAB, LabVIEW, Python, C, Data Acquisition (DAQ), Signal Processing

Work Experience

Engineering Product Innovation Center (E.P.I.C.), Boston, MA.

May 2024 - Present

Laboratory Assistant

- Supported prototyping and fabrication for engineering projects in a high-traffic makerspace
- Operated CNC mills, CNC lathes, manual mills/lathes, and 3D printers (*FFF* and *SLA*)
- Generated CAM toolpaths and G-code using SolidWorks and Autodesk Fusion 360
- Assisted with part fixturing, tolerancing, and design adjustments for manufacturability
- Enforced lab safety and proper machine operation while supporting multiple users

Starbucks, Breezewood, PA and Boston, MA

Jun. 2021 – Sep. 2025

Barista

- Balanced full-time Mechanical Engineering coursework with part-time employment
- Worked in a fast-paced, team-based environment requiring accuracy, multitasking, and efficiency

Projects

Reactive Road Sign System – Senior Design I & II (*ME460/461*)

- Designed a retrofit mechanical system for roadside signposts to deflect under extreme wind loads
- Evaluated joint concepts using CAD, materials selection, and structural failure analysis
- Modeled wind-induced loading and bending moments to size a sacrificial breakaway pin
- Built and tested physical prototypes to validate loading behavior and manufacturability

Compact Fishing Kit Product – Automation & Manufacturing Methods (*ME345*)

- Designed a consumer product and developed a CAD-to-CAM workflow for CNC manufacturing
- Created CNC toolpaths and programmed collaborative robots for machine tending and assembly
- Integrated conveyors, robot programs, and CNC machines into an automated manufacturing cell

Electromechanical Interactive Ouija Board – Electromechanical Design (*ME360*)

- Built a 2-axis, belt-driven motion system to position a planchette using stepper motors
- Implemented motion control to convert AI-generated responses into coordinate targets
- Developed control software integrating voice input, AI API communication, and motor control

Adaptive LED Lamp – Introduction to Engineering Design (*EK210*)

- Designed and built a motion-activated LED lamp using electronic sensing and physical prototyping
- Implemented ambient-light sensing and time-based control to reduce blue-light exposure